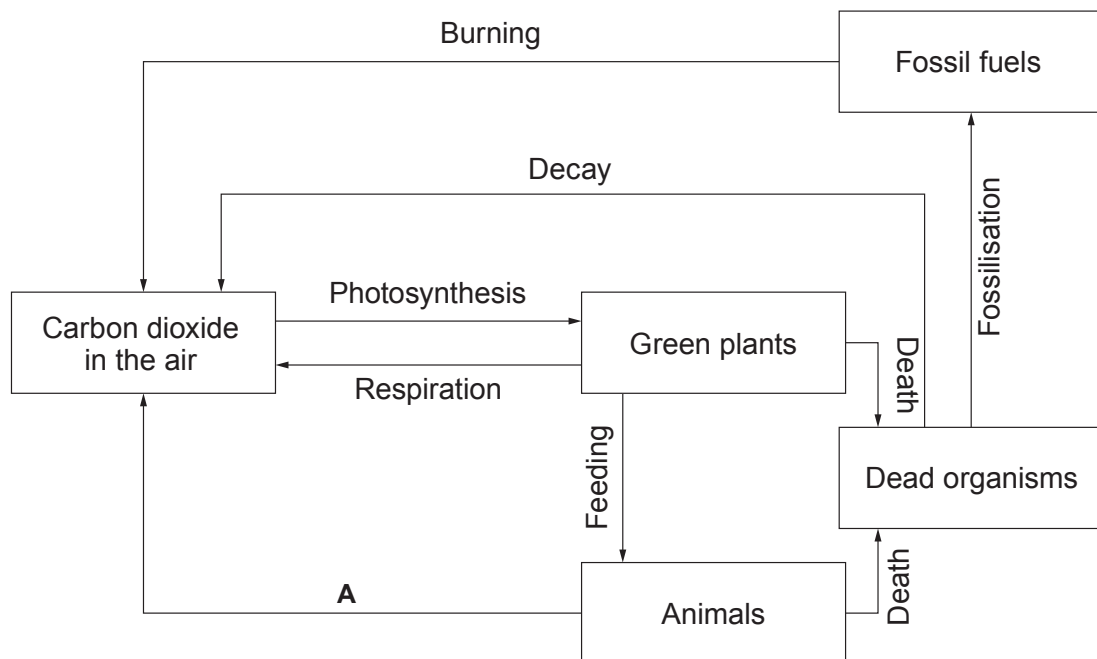


2. The diagram shows the carbon cycle.



(a) Use the diagram and your knowledge to answer the following questions.

(i) Circle the name of the process shown by label **A**.

[1]

feeding photosynthesis respiration

(ii) Tick (✓) the boxes next to the **three** correct statements.

[3]

Photosynthesis uses carbon dioxide from the atmosphere.

☐

Decomposing dead plants adds carbon dioxide to the atmosphere.

☐

Fossilising dead plants adds carbon dioxide to the atmosphere.

☐

Respiration adds carbon dioxide to the atmosphere.

☐

Burning dead plants removes carbon dioxide from the atmosphere.

☐

When green plants feed they add carbon dioxide to the atmosphere.

☐

(iii) State **two** ways that human activity has caused carbon dioxide levels to increase.

[2]

1.

2.



- (b) The table shown below is taken from a climate change website.

It gives the mass of carbon dioxide produced by a person travelling to work either by car or bus.

Distance travelled (km)	20
Mass of carbon dioxide produced per person travelling by car (kg)	1.20
Mass of carbon dioxide produced per person travelling by bus (kg)	0.20

Four people each travel a total of 20 km per day.
Use the information to answer the following questions.

- (i) Use the equation below to answer the following questions.

total mass of carbon dioxide produced (kg) = number of people $\times$ mass of carbon dioxide produced per person

- I. Calculate the total mass of carbon dioxide produced when four people drive 20 km by car. [2]

mass of carbon dioxide produced = kg

- II. Calculate the mass of carbon dioxide produced if four people travelled 20 km by bus. [1]

mass of carbon dioxide produced = kg

- (ii) Calculate the savings in carbon dioxide production per week if they travel by bus on the same 20 km journey from Monday to Friday instead of using their cars. [2]

savings in carbon dioxide production = kg

